CHAPTER FOUR

Thermal science and engineering of β-Ga_2O_3 materials and devices

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1. Crystal structure and thermal conductivity of bulk β-Ga_2O_3

Gallium oxide is not a new material and its five polymorphs were reported back to 1952 (Roy et al., 1952), which are: δ-Ga_2O_3, ε-Ga_2O_3, α-Ga_2O_3, β-Ga_2O_3, and γ-Ga_2O_3. The β-phase gallium oxide (β-Ga_2O_3) has the most stable crystal structure among the five commonly identified polymorphs of Ga_2O_3 (Pearton et al., 2018a,b). β-Ga_2O_3 has a melting temperature above 1700 °C, and other polymorphs can convert into β-Ga_2O_3 at higher temperatures and the detailed transformation relationship and conditions are well-documented in literature (Roy et al., 1952). The crystal system of β-Ga_2O_3 is monoclinic with a primitive cell of 10 atoms, 4 gallium, and 6 oxygen, with 1 gallium ion in tetrahedral and another in octahedral sites as shown in Fig. 1A. Peelaers et al. studied the Brillouin zone and the band structure of β-Ga_2O_3 (Peelaers and Van de Walle, 2015;

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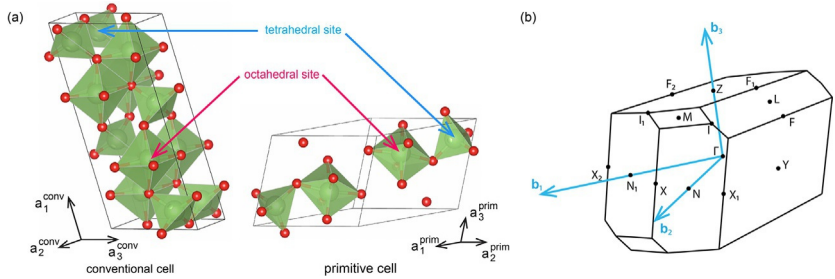


Fig. 1 (A) Conventional unit cell and primitive cell of β-Ga_2O_3 (Peelaers and Van de Walle, 2015). (B) First Brillouin zone and high-symmetry points of β-Ga_2O_3 (Peelaers and Van de Walle, 2015).

Peelaers et al., 2018). The relations between the primitive translation vectors and the conventional unit cell (with 20 atoms) vectors are shown below:

$$\boldsymbol{a}_1^{prim} = \frac{\left(\boldsymbol{a}_1^{conv} + \boldsymbol{a}_2^{conv}\right)}{2}, \boldsymbol{a}_2^{prim} = \frac{\left(-\boldsymbol{a}_1^{conv} + \boldsymbol{a}_2^{conv}\right)}{2}, \boldsymbol{a}_3^{prim} = \boldsymbol{a}_3^{conv}.$$

With the accurately defined β-Ga_2O_3 primitive cell, the Brillouin zone can be obtained as the Wigner-Seitz cell of the reciprocal lattice as shown in Fig. 1B (Peelaers and Van de Walle, 2015). It is important to use the Wigner-Seitz cell as the Brillouin zone of the primitive cell instead of the parallelepiped with reciprocal vectors of either the primitive cell or the conventional unit cell to accurately capture high-symmetry points with the whole irreducible Brillouin zone (Peelaers and Van de Walle, 2015). Advanced hybrid functional was used to capture the lattice length and electron band structure (Peelaers et al., 2018). The experimental (calculated) lattice parameters and direct bandgap are:

$$a = 1.221\ (1.227)\ \text{nm}, b = 0.304\ (0.305)\ \text{nm}, c = 0.580\ (0.582)\ \text{nm}$$
$$\beta = 103.83°(103.82°), E_{gap} = 4.76\ (4.88)\ \text{eV},$$

Due to the development of the growth techniques and potential applications such as power electronics and solar-blind UV detectors, β-Ga_2O_3 has attracted great attention recently (Pearton et al., 2018a,b; Tsao et al., 2018). The wafers grown from the melt have the potential to lower the cost compared to the sublimation techniques (Pearton et al., 2018a). It enables the growth of large-scale affordable high-quality bulk crystals which is one of the main driving forces for electronics and optoelectronics applications (Pearton et al., 2018a). Recently, metal organic chemical vapor deposition (MOCVD) has also demonstrated epitaxial films with record-high carrier mobility (Feng et al., 2019; Zhang et al., 2019). A max DC power

density of about 60 W/mm was achieved for a field-effect transistor (FET) fabricated by a mechanically exfoliated β-Ga_2O_3 thin film transferred on single crystal diamond, which is the highest power density demonstrated by any semiconductor device (Tadjer et al., 2019). All these works show the great potential of β-Ga_2O_3 for applications of power electronics due to its ultrawide bandgap (4.8 eV) and high critical electric field (8 MV/cm), resulting in a Baliga figure of merit more than 3000 times higher than that of Si (Higashiwaki and Jessen, 2018; Higashiwaki et al., 2012).

However, a main challenge for β-Ga_2O_3 applied in power electronics is its low and strongly anisotropic thermal conductivity (Cheng et al., 2019a; Guo et al., 2015). The localized Joule-heating increases the channel temperature and degrades the device performance (Cheng et al., 2018, 2019b). Due to the complex crystal structure, there are 10 atoms in 1 unit cell, resulting in 30 phonon branches (3 acoustic and 27 optical) in the phonon dispersion relation of β-Ga_2O_3. The strong phonon-phonon scatterings lead to a low thermal conductivity of β-Ga_2O_3 (Santia et al., 2015). As shown in Fig. 2, the thermal conductivity of β-Ga_2O_3 is at least one order of magnitude lower than other wide bandgap semiconductors such as GaN and SiC because of the strong phonon-phonon scattering (Cheng et al., 2020a; Tsao et al., 2018). Moreover, the thermal conductivity of β-Ga_2O_3 along the [100] direction is more than twice lower than that along the [010] direction, showing strong anisotropy (Guo et al., 2015; Jiang et al., 2018).

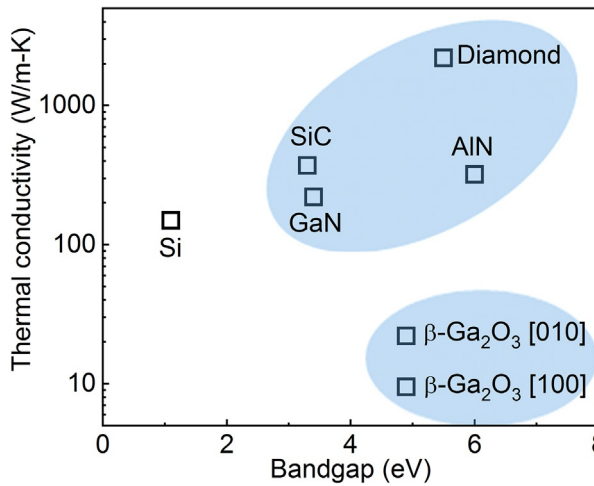


Fig. 2 Thermal conductivity of wide/ultrawide bandgap semiconductors at room temperature (Cheng et al., 2020a; Tsao et al., 2018). Silicon is also included as reference.

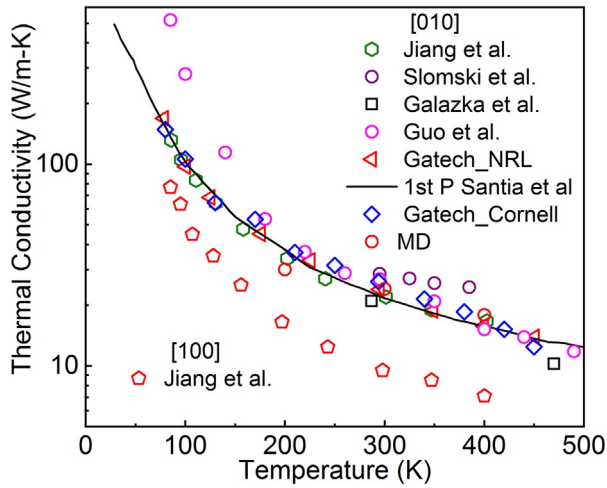


Fig. 3 Temperature dependent thermal conductivity of bulk β-Ga_2O_3 with different orientations (Aller et al., 2018; Blumenschein et al., 2018; Cheng et al., 2019a; Galazka et al., 2014; Guo et al., 2015; Handwerg et al., 2015; Jiang et al., 2018; Santia et al., 2015). *MD*, molecular dynamics simulation; *1st P*, first-principle calculation. All the other data are experimental measurements.

Despite the importance of necessary aggressive thermal management strategy for β-Ga_2O_3-based power electronics, the thermal study on β-Ga_2O_3 is disproportionately less than the demonstrations of devices (Reese et al., 2019). Fig. 3 shows typical experimental measurements of temperature dependent thermal conductivity of β-Ga_2O_3 along different orientations (Blumenschein et al., 2018; Cheng et al., 2019a; Galazka et al., 2014; Guo et al., 2015; Handwerg et al., 2015; Jiang et al., 2018; Santia et al., 2015). In the temperature range studied in Fig. 3, the thermal conductivity of β-Ga_2O_3 decreases with increasing temperature. The variation of different measurements on the thermal conductivity of β-Ga_2O_3 along [010] direction should be due to difference in sample quality and measurement accuracy. As temperature increases, the number and energy of excited phonons increases, leading to increasingly extensive phonon-phonon scatterings and reduced thermal conductivity. The thermal conductivity along the [100] direction is also included to show the temperature dependent anisotropic thermal conductivity, which is much lower than that along [010] direction. Some of current devices are grown on [010] substrates which takes the advantage of the relatively high thermal conductivity along this direction. For high temperature applications, aggressive thermal management strategies are essential due to the further reduced thermal conductivity.

The thermal conductivity of pure bulk β-Ga_2O_3 can be calculated from Boltzmann transport equation (BTE) together with phonon-phonon scattering rates obtained from the 2nd and 3rd order force constants generated with first-principle calculations. Density-functional-theory (DFT) calculations with Quantum ESPRESSO (Giannozzi et al., 2009) were performed to obtain the structure and both harmonic and anharmonic properties of β-Ga_2O_3 (Santia et al., 2015). Phonopy (Togo and Tanaka, 2015) was used to calculate the 2nd order interatomic force constants and ShengBTE (Li et al., 2014) was used to obtain the 3rd order force constants by solving BTE. At room temperature, the calculated thermal conductivity (κ) along the [100] and [010] directions are 16.06 and 21.54 W/mK while their experimental counterparts are 13 and 21 W/mK. The temperature dependent thermal conductivity in the [010] orientation based first-principle calculations are shown in Fig. 3. The strong anisotropy comes from the monoclinic lattice structure of β-Ga_2O_3. However, the calculated thermal conductivity along the [001] direction does not agree with measurements, which was attributed to additional scattering mechanisms in the measured samples (Santia et al., 2015). It should be noted that small supercell might lead to the calculation inaccuracy by considering a $2 \times 2 \times 2$ supercell for the 2nd and 3rd order force constants calculation was used. The contribution of optical phonons is important for β-Ga_2O_3 especially for the highest thermal conductivity orientation [010] with about 44% optical contribution (Santia et al., 2015).

Equilibrium molecular dynamics (EMD) method was applied to simulate the thermal conductivity of α-Ga_2O_3, β-Ga_2O_3, and ε-Ga_2O_3 at room temperature with different orientations (Aller et al., 2018). The temperature dependent thermal conductivity of β-Ga_2O_3 were also calculated and the temperature dependent thermal conductivity of β-Ga_2O_3 in the [010] orientation are shown in Fig. 3 to compare with experimental results (Aller et al., 2018). The interatomic potential applied in MD simulation is a rigid-ion model (Blanco et al., 2005), which studied the point defects of β-Ga_2O_3 with lattice dynamics (LD) method. Both the structural parameters and thermal conductivity can be accurately simulated from EMD. The lattice parameters from MD minimization are:

$$a = 1.22 \text{ nm}, b = 0.314 \text{ nm}, c = 0.58 \text{ nm}, \beta = 103.7°,$$

which are consistent with experimental measurements. β-Ga_2O_3 has the strongest anisotropy among gallium oxide polymorphs due to its strongest anisotropy in crystal structure. The thermal conductivity of ε-Ga_2O_3 is

much lower than that of α-Ga_2O_3 and β-Ga_2O_3. It should be noted that there might be some inaccuracy related to quantum effect since MD is classical together with the high Debye temperature of Ga_2O_3. However, the consistency between EMD results and experimental measurements suggests that quantum effect might be negligible for β-Ga_2O_3 at least above 200 K (Aller et al., 2018).

2. Thermal conductivity of β-Ga_2O_3 nanostructures

Compared with the low thermal conductivity of bulk β-Ga_2O_3, the thermal conductivity of β-Ga_2O_3 nanostructures is further reduced due to the size effect from phonon-imperfection scatterings, such as film boundaries or alloy. For device applications, these nanostructures compose of the key components of the devices. For instance, β-Ga_2O_3/$(Al_{0.1}Ga_{0.9})_2O_3$ superlattices were used to generate two-dimensional electron gas, similar to GaN/AlGaN structures (Zhang et al., 2018a,b). Fig. 4A shows a cross-section image of β-Ga_2O_3/$(Al_{0.1}Ga_{0.9})_2O_3$ superlattices while the Fig. 4B shows its temperature dependent thermal conductivity (Cheng et al., 2019a). At room temperature, a 5.7-times reduction in thermal conductivity was observed for the superlattices compared with the bulk counterpart (Cheng et al., 2019a). The superlattice thermal conductivity (4.6 W/m-K) is reduced twice even comparing with the bulk thermal conductivity along the [100] direction (Cheng et al., 2019a). The reduction increases as temperature decreases. For bulk β-Ga_2O_3, phonon-phonon scatterings become much less extensive as temperature decreases, leading to increased thermal conductivity. While for superlattices, phonon-boundary and phonon-alloy scatterings dominate at low temperatures, leading to decreased thermal conductivity. That's why the measured thermal conductivity of the bulk and the superlattice show different trends for temperature dependence, as shown in Fig. 4B. For high temperatures above 380 K, the thermal conductivity of both bulk and superlattices decreases with increasing temperature because phonons are fully excited and the phonon-phonon scatterings become increasingly extensive.

Alloy usually lowers thermal conductivity. However, a recent calculation showed that the thermal conductivity of ordered β-$(Al_xGa_{1-x})_2O_3$ can be higher than pure β-Ga_2O_3 (Mu et al., 2019a). For β-$(Al_{0.5}Ga_{0.5})_2O_3$ alloy, phonon group velocity is higher than that of β-Ga_2O_3 and phonon-phonon scattering rates are smaller than that of β-Ga_2O_3, leading to a 70% increase in thermal conductivity compared with β-Ga_2O_3 (Mu et al., 2019a). Here, the

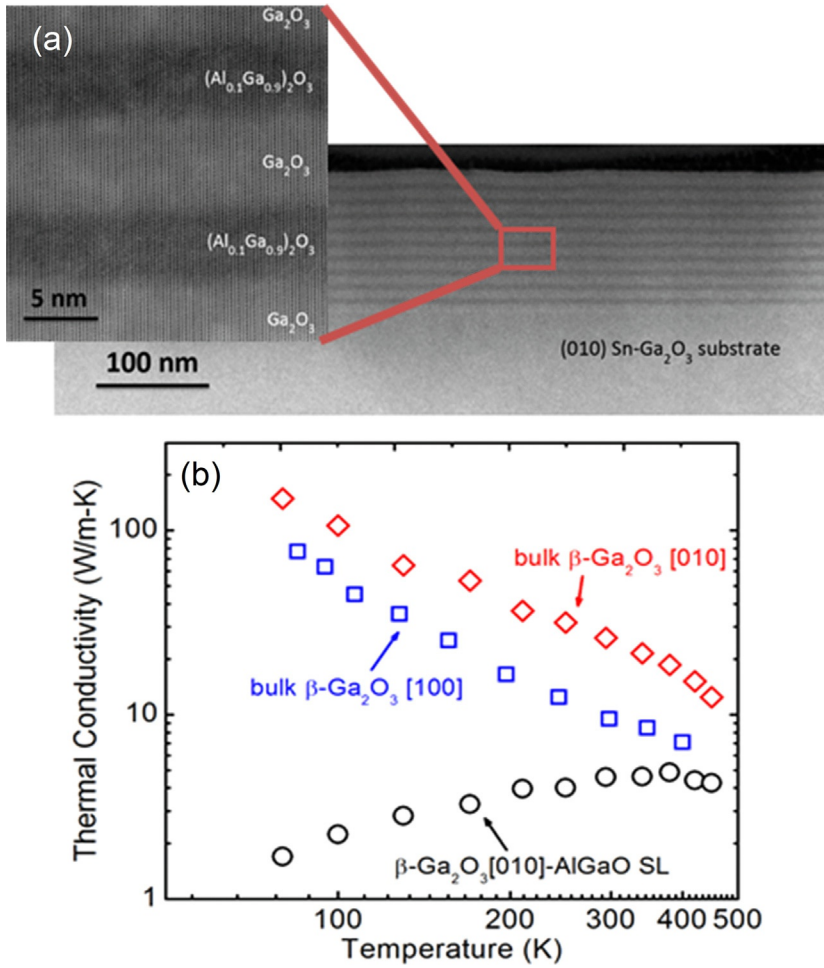


Fig. 4 (A) TEM image of the cross-section of β-Ga_2O_3/AlGaO superlattices. (B) Temperature dependence of significant reduction in thermal conductivity of β-Ga_2O_3/AlGaO superlattices (Cheng et al., 2019a; Jiang et al., 2018). Panel A*: Copyright permission from Cheng, Z., et al., 2019a. Significantly reduced thermal conductivity in beta-(Al0.1Ga0.9) 2O3/Ga2O3 superlattices. Appl. Phys. Lett. 115, 092105.*

Al ion is always at the octahedral site in the unit cell, because Al ions prefer the octahedral sites compared to the tetrahedral sites from DFT with accurate hybrid functional (Mu et al., 2019a). The ordered β-$(Al_{0.5}Ga_{1-0.5})_2O_3$ lowers phonon scattering rates, unlike disordered alloy. The primitive cell of β-$(Al_{0.5}Ga_{0.5})_2O_3$ has the same number of atoms as the pure β-Ga_2O_3, but with a smaller atomic mass, and as a result the phonon cut-off frequency

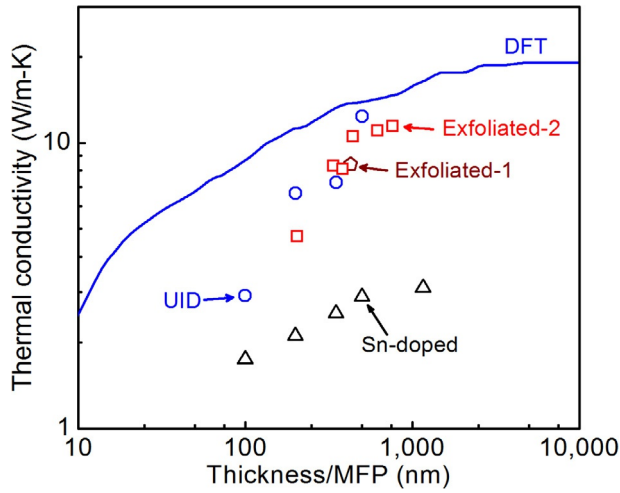


Fig. 5 Thickness dependent thermal conductivity of β-Ga_2O_3 perpendicular to the (-201) plane. "UID" represents the thermal conductivity of unintentionally doped β-Ga_2O_3 thin films while "Sn-doped" represents that of the Sn-doped β-Ga_2O_3 thin films (Blumenschein et al., 2018). "Exfoliated-1" and "Exfoliated-2" represent the thermal conductivity of β-Ga_2O_3 thin films mechanically exfoliated from bulk crystals along (-201) plane (Cheng et al., 2019c; Zhang et al., 2020). MFP-accumulated thermal conductivity is also included as comparison, which is calculated by DFT in this work.

and group velocity is higher than that of β-Ga_2O_3 (Mu et al., 2019a). A reduced overlap between acoustic and optical branches was also observed, resulting in a reduced phonon-phonon scattering rates and contribute to the increased thermal conductivity (Mu et al., 2019a).

Fig. 5 provides a summary of the thickness dependent thermal conductivity of β-Ga_2O_3 perpendicular to the (-201) plane. The thermal conductivity of β-Ga_2O_3 thin films with different doping levels and fabrication methods are compared: unintentionally doped (UID), Sn-doped, and mechanically exfoliated (Blumenschein et al., 2018; Cheng et al., 2019c; Zhang et al., 2020). Mean free path (MFP) accumulated thermal conductivity is also included as comparison, which is calculated by DFT in this work. We can observe strong reduction in thermal conductivity because the film boundary scatters with phonons and limits phonon mean free path, leading to reduced thermal conductivity. The mechanically exfoliated thin films have bulk-like quality because they are peeled off from bulk crystals. The UID thin films have similar thickness dependent thermal conductivity trends with the exfoliated thin films, showing similar quality to bulk crystals. The deviation from the measured thermal conductivity from the calculated

MFP-accumulated thermal conductivity shows the quasi-ballistic thermal conduction in these thin films. For phonons with MFP shorter than the film thickness, they have some probability to scatter with the film boundaries as well. This derivation increases with decreasing thickness. As thickness decreases, an increasing number of phonons are limited by the film boundaries and ballistic or quasi-ballistic thermal transport dominates thermal conduction in these thin films. Additionally, for the Sn-doped thin films, the thermal conductivity reduces further due to additional phonon scattering sources (phonon-defect scatterings). Please note that doping is necessary for some components in device fabrications, which will further bring down their thermal conductivity. As shown in Fig. 5, Sn-doping is able to deteriorate thermal conduction several times compared with UID films (Blumenschein et al., 2018).

3. Heterogeneous integration with diamond

Because of the low thermal conductivity of bulk β-Ga_2O_3 and further reduced thermal conductivity of β-Ga_2O_3 nanostructures which ware necessary in devices, aggressive thermal management strategies are essential for stable device performance and good reliability (Reese et al., 2019). To cool GaN-based high electron mobility transistor (HEMTs), diamond was integrated with GaN to take advantage of the high thermal conductivity of diamond (Anaya et al., 2017; Cheng et al., 2018, 2019b, 2020a; Yates et al., 2016). Similarly, diamond can be used to cool β-Ga_2O_3 based devices as well (Cheng et al., 2019c, 2020b). Fig. 6 shows the integration of β-Ga_2O_3 with single crystal diamond. Fig. 6A shows the atomic force microscope (AFM) image of the mechanically exfoliated β-Ga_2O_3 thin films transferred to single crystal diamond via van der Waals bonding while Fig. 6B shows the cross-section TEM image of the nanocrystalline β-Ga_2O_3 thin films grown on single crystal diamond substrates via atomic layer deposition (ALD). The devices based on the transferred nanoscale membranes have demonstrated record-high power density, which shows the great potential of β-Ga_2O_3 materials but this method cannot be scaled up for wafer-scale fabrications. The ALD-deposited β-Ga_2O_3 are nanocrystalline which are difficult to fabricate devices but can be used as transition layers for further growth of β-Ga_2O_3. No voids were observed at the ALD-deposited β-Ga_2O_3-diamond interfaces (Cheng et al., 2020b).

The thermal boundary conductance (TBC) of β-Ga_2O_3-diamond interfaces were measured by time-domain thermoreflectance (TDTR). Fig. 7

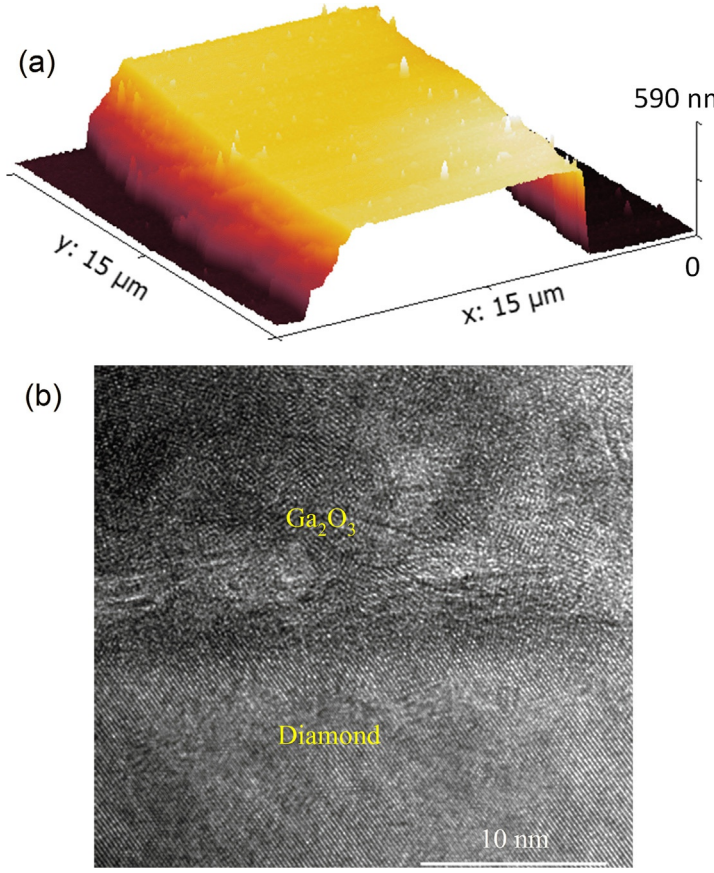


Fig. 6 Integration of β-Ga_2O_3 with single crystal diamond. (A) The AFM image of the mechanically exfoliated β-Ga_2O_3 thin films transferred to single crystal diamond via van der Waals bonding. (B) The cross-section TEM image of the nanocrystalline β-Ga_2O_3 thin films grown on single crystal diamond substrates via ALD. *Reproduction permissions from Cheng, Z., et al., 2019c. Thermal conductance across β-Ga2O3-diamond van der Waals heterogeneous interfaces. APL Mater. 7, 031118; Cheng, Z., et al., 2020b. Integration of polycrystalline Ga2O3 on diamond for thermal management. Appl. Phys. Lett. 116, 062105.*

shows the TBC of diamond related interfaces fabricated by different methods: transferred van der Waals bonding, physical vapor deposition (PVD), chemical vapor deposition (CVD), and ALD (Cheng et al., 2019b,c, 2020b; Lyeo and Cahill, 2006; Oh et al., 2011; Zhang et al., 2020). The transferred Au-Si/SiO_2/Al_2O_3 interfaces are also included for comparison (Oh et al., 2011). The TBC of the transferred β-Ga_2O_3-diamond interfaces are about 17 MW/m^2-K, close to other transferred interfaces

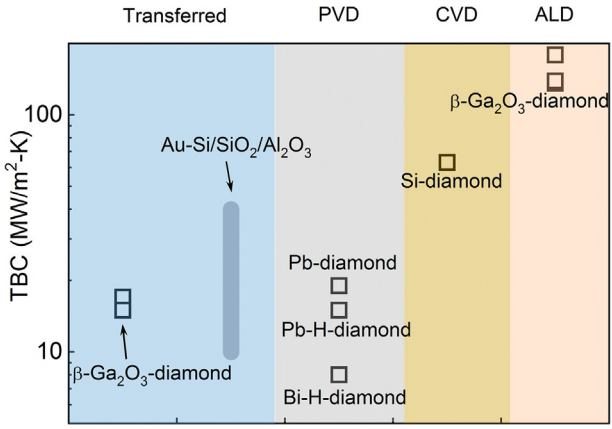


Fig. 7 TBC of diamond related interfaces fabricated by different methods: transferred van der Waals bonding, PVD, CVD, and ALD (Cheng et al., 2019b,c, 2020b; Lyeo and Cahill, 2006; Oh et al., 2011; Zhang et al., 2020). The transferred Au-Si/SiO_2/Al_2O_3 interfaces are also included for comparison (Oh et al., 2011).

(Cheng et al., 2019c; Oh et al., 2011; Zhang et al., 2020). The TBC across β-Ga_2O_3-diamond interfaces can be theoretically predicted with Landauer approach with transmission function from diffuse mismatch model (DMM) (Cheng et al., 2019c; Oh et al., 2011; Zhang et al., 2020). The expression of the Landauer formula is:

$$G = \sum_p \frac{1}{2} \iint D_1(\omega) \frac{df_{BE}}{dT} \hbar\omega v_1(\omega) \tau_{12}(\theta, \omega) \cos\theta \sin\theta \; d\theta \; d\omega, \qquad (1)$$

where D is the phonon DOS, f_{BE} is the Bose-Einstein distribution function, $\hbar$ is the reduced Planck constant, ω is the phonon angular frequency, v is the phonon group velocity of material 1 (Ga_2O_3), τ_{12} is the transmission coefficient from material 1 to 2 (here it is from Ga_2O_3 to diamond), θ is the angle of incidence, and the sum is over all incident phonon modes. The transmission function from DMM is:

$$\tau_{12}(\omega) = \frac{\sum_p M_2(\omega)}{\sum_p M_1(\omega) + \sum_p M_2(\omega)}, \qquad (2)$$

where M is the phonon number of modes (Cheng et al., 2019c; Oh et al., 2011; Zhang et al., 2020). The calculated TBC from the Landauer approach across β-Ga_2O_3-diamond interface at room temperature is 312 MW/m^2K,

which is larger than the experimental measurements. The reason for the large difference is that bonding at the exfoliated β-Ga_2O_3-diamond interface is van der Waals and is very weak compared with covalent bonds. The phonon transmission at weakly-bonded interface is very low. Moreover, the interfacial contact area is also very small compared with deposited or grown interfaces. As a result, the measured TBC is low.

For metals with low Debye temperatures such Pb and Bi, metal-diamond TBC are comparable to transferred β-Ga_2O_3-diamond interfaces even if they are deposited by PVD which supposed to have excellent interface contacts. TBC of Si-diamond interfaces grown by CVD and β-Ga_2O_3-diamond interfaces grown by ALD have larger values than those of the interfaces mentioned above due to good interface contact and good phonon density of states matching (Cheng et al., 2019b, 2020b). The TBC values impact heat dissipation significantly because interfaces are close to Joule-heating hot-spots. The thermal boundary resistance accounts for a large portion of the total thermal resistances (Cheng et al., 2019b,c, 2020a).

To estimate the effect of the β-Ga_2O_3–substrate TBC on the temperature rises of a power device, an analytical solution for the temperature rise calculation in multilayer structures with discrete heat sources were used (Bagnall et al., 2014). A 500-nm (100) Ga_2O_3 layer atop a substrate has anisotropic thermal conductivity with $k_z = 12\,W/m\text{-}K$ and $k_r = 21\,W/m\text{-}K$ (Santia et al., 2015). The modeled device structure and conditions were shown in Fig. 8A: 10 fingers with 50 μm gate-to-gate spacing, $4 \times 150\,\mu m$ for each heat source, $2000 \times 2000\,\mu m$ total domain, and 10 W/mm total power density. The max device temperature with different TBC and different substrates (Si, SiC, and diamond) were calculated and the results are shown in Fig. 8B (Cheng et al., 2019c). Compared with Si substrates, the max temperature of devices with high thermal conductivity substrates such as SiC and diamond are much lower for a certain TBC values. For a certain substrate, TBC plays an important role in limiting the max device temperature when TBC value is lower than $100\,MW/m^2\text{-}K$ for the modeled device structures in this work. For single crystal diamond substrate with a relatively low TBC of the transferred β-Ga_2O_3-diamond interfaces, the cooling performance is excellent, achieving a record-high power density (Tadjer et al., 2019). For TBC higher than $200\,MW/m^2\text{-}K$, the max device temperatures are weakly dependent on TBC values for the modeled device structures in this work.

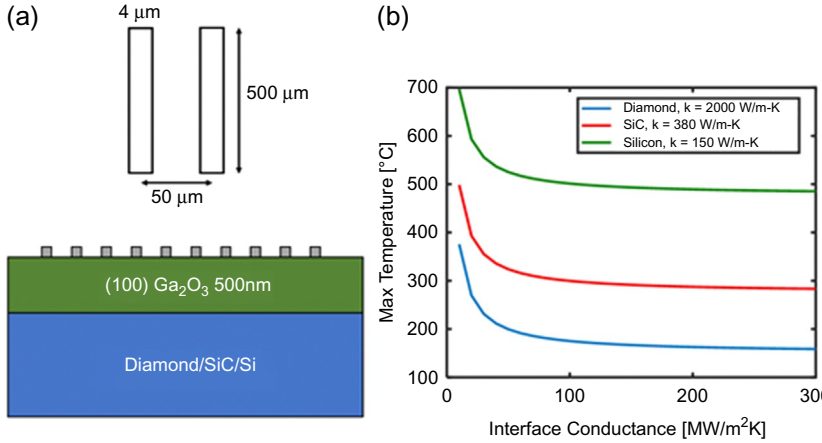


Fig. 8 Effect of β-Ga_2O_3–substrate TBC on device thermal management. (A) Schematic diagram of device structure. (B) Effect of β-Ga_2O_3–substrate TBC on the max temperature of a device (Cheng et al., 2019c). *Reproduction permissions from Cheng, Z., et al., 2019c. Thermal conductance across β-Ga2O3-diamond van der Waals heterogeneous interfaces. APL Mater. 7, 031118.*

4. Wafer bonding of monocrystalline β-Ga_2O_3 on SiC

For scalable fabrication of β-Ga_2O_3 devices on high thermal conductivity substrates, an excellent technique is surface activated bonding, which have been proved to be able to bond wafers at room temperature with high TBC (Cheng et al., 2020a; Mu et al., 2019b). Recently, β-Ga_2O_3 was directly bonded on 4H-SiC via surface activated bonding (Xu et al., 2019). Wafer-scale integration of monocrystalline β-Ga_2O_3 thin films on 4H-SiC wafers has also been demonstrated via surface activated bonding technique and smart-cut technique (Cheng et al., 2020c). Hydrogen ions are implanted into β-Ga_2O_3 wafers before bonded to SiC wafers. With a low-temperature annealing process, the ions propagate into a certain layer under the β-Ga_2O_3 wafer surface and exfoliate a layer of β-Ga_2O_3 thin film from the β-Ga_2O_3 wafer. The transferred β-Ga_2O_3 thin film on the SiC can be polished to obtain a smooth surface and the β-Ga_2O_3 wafer can be used repeatedly for exfoliation (Cheng et al., 2020c). The measured TBC of these bonded interfaces are reasonable high enough to achieve good cooling performance (Cheng et al., 2020c). This work may lay the foundation of

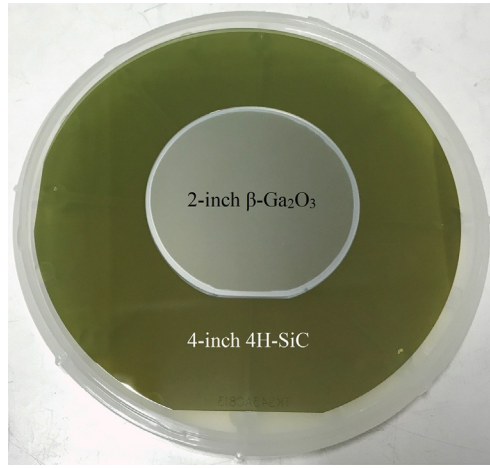


Fig. 9 A 2-in. β-Ga_2O_3 wafer is bonded with a 4-in. 4H-SiC wafer by the surface activated bonding technique at room temperature. *Courtesy by Prof. Fengwen Mu.*

large-scale production of β-Ga_2O_3 devices with excellent thermal management to cool the Joule-heating in power electronics. Similar work can be done to bond β-Ga_2O_3 with single crystal diamond substrates as well. The smart-cut monocrystalline β-Ga_2O_3 thin films have bulk-like quality without grain boundaries or large amount of defects after a high temperature annealing (for instance 800 °C). This will enable homo-epitaxial growth of high performance β-Ga_2O_3 devices on these smart-cut wafers. Fig. 9 shows a 2-in. β-Ga_2O_3 wafer bonded with a 4-in. 4H-SiC wafer by the surface activated bonding technique at room temperature. No visible voids are observed at the bonded interfaces. For more details about the smart-cut and bonding processes can be found in the reference (Cheng et al., 2020c).

5. Device-level cooling

Regarding the device level, any processes including the thinning, doping or heterogeneous growth, may diminish the phonon transport and hence the thermal conductivity. The low thermal conductivities of β-Ga_2O_3 and its related nanostructures in devices lead to the large intrinsic thermal resistance of the devices. For example, a thermal resistance of 88 mm °C/W has been measured for a typical β-Ga_2O_3 metal-oxide-semiconductor field-effect transistors (MOSFETs), 6 × larger than the multifinger GaN high electron mobility transistors (HEMTs) with a similar gate width and active heat source separation on the high thermal conductivity

SiC (420 W/mK) substrate, and 2 × larger than GaN HEMTs on a low thermal conductivity sapphire (24 W/mK) (Pomeroy et al., 2018). The large intrinsic thermal resistance may cause temperature induced reliability issues, with the severe examples of thermal droop (Singh et al., 2018) and catastrophic failure (Green et al., 2017) observed in β-Ga_2O_3 MOSFETs under RF-operation with high drain bias. Therefore, increasing attention has been paid to the thermal management of β-Ga_2O_3 devices.

Over the past decade, a number of die-level thermal management approaches have been developed for GaN HEMTs, which have allowed them to reach operational powers in laboratory testing. Some of the approaches include heterogeneous integration with high thermal conductivity substrates (SiC and diamond (Cheng et al., 2020a; Cho et al., 2012; Francis et al., 2010; Mandal et al., 2019; Mu et al., 2019b; Pomeroy et al., 2014)), flip-chip packaging for top side heat extraction (Das et al., 2006; Ji et al., 2006), or overgrowth of nanocrystalline diamond heat spreaders on top of the device layers for heat spreading (Yates et al., 2018; Zhou et al., 2017a). Other approaches (Al-Neama et al., 2018; Ditri et al., 2015) have involved the use of liquid cooling to provide high convective heat transfer coefficients and reduce the overall junction temperature. Regardless of the approach, the key is to move high performance cooling solutions close to the heat generating region of the device and, thus, reduce the overall thermal resistance of the device architecture. As the β-Ga_2O_3 lateral field-effect transistors (FETs) structure is similar to GaN HEMTs structure, some device-level cooling methods were designed (Chatterjee et al., 2019; Pomeroy et al., 2018; Yuan et al., 2020), inheriting the existing expertise developed for GaN HEMTs, and the electric-thermal simulations were conducted to evaluate the methods' cooling performance. Though the devices studied in references (Chatterjee et al., 2019; Pomeroy et al., 2018; Yuan et al., 2020) are somehow different in terms of device layout, β-Ga_2O_3 substrate orientation, etc., the cooling methods proposed in them feature some basic forms of architecture, for example, the bottom side cooling, top side cooling, and double side cooling, as shown in Fig. 10A–C. The electric-thermal simulations combine a 2-D drift diffusion model, achieved in Silvaco ATLAS or Synopsys Sentaurus, for the Joule-heating distribution calculation, which is then input into a 3-D finite element (FE) thermal model (using ANSYS or COMSOL) to calculate the device temperature.

The bottom side cooling method integrates the Ga_2O_3 material stack with a high thermal conductivity heterogeneous substrate, followed by a

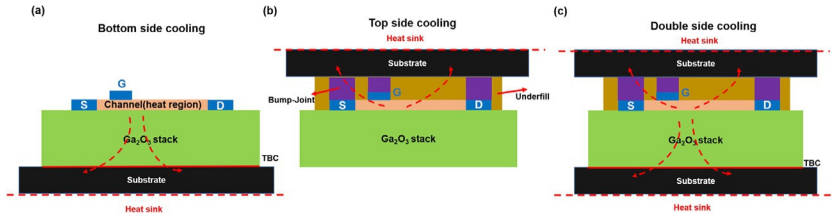


Fig. 10 Schematic of three basic cooling schemes implemented for a β-Ga_2O_3 lateral field-effect transistors (FETs): (A) bottom side cooling; (B) top side cooling; and (C) double side cooling.

heat sink, as shown in Fig. 8A. The heat sink could be several different ways such as an additional package component, microchannel module, or an isothermal boundary with a fixed temperature (Chatterjee et al., 2019; Pomeroy et al., 2018; Yuan et al., 2020). The generated heat is dissipated into the heat sink through the low thermal conductivity Ga_2O_3 layer, the Ga_2O_3/substrate thermal interface, and the substrate. Thus, the effect of Ga_2O_3 thickness (t_g), thermal boundary conductance between Ga_2O_3 and the substrate (TBC) and substrate thermal conductivity (k_s) were investigated in the studies (Chatterjee et al., 2019; Pomeroy et al., 2018; Yuan et al., 2020). For the MESFET device process on (010) Ga_2O_3, thinning the Ga_2O_3 from 100 to 1 μm results in a ~50% reduction in channel maximum temperature (T_{max}) (Yuan et al., 2020). For the MOSFET device on (010) Ga_2O_3, a ~75% reduction was achieved when reducing t_g from 600 to 6 μm (Pomeroy et al., 2018), while a 3–4 times reduction was observed for the MOSFET device on (-201) Ga_2O_3 when thinning the Ga_2O_3 from 100 to 10 μm. Though thinning Ga_2O_3 contributes to a benefit (T_{max} reduction) with a different quantity in those studies, likely due to the differences in device structures (i.e., layout, β-Ga_2O_3 substrate orientation) and other extrinsic factors (such as TBC, substrate and heat sink), all results demonstrate a significantly great impact of t_g on T_{max}. This is owing to the fact that the device thermal resistance is much sensitive to the thickness given that the thermal conductivity of Ga_2O_3 in the thickness direction is much low. It is found that increasing TBC reduces T_{max}, and the TBC effect is highly dependent on the Ga_2O_3 thickness (t_g) that TBC shows a greater impact on T_{max} as t_g decreases (Yuan et al., 2020). The reason is that for the thick layer of Ga_2O_3, the Ga_2O_3 layer thermal resistance (R_G) dominates the device total thermal, whereas for the thin Ga_2O_3 layer, the thermal resistance at the interface (R_B) become comparable to R_G and begins to contribute to the temperature rise. T_{max} reduces when k_s is increasing (Chatterjee et al., 2019; Yuan et al., 2020). It is also revealed in Yuan et al. (2020) that after

k_s exceeds some value (i.e., 400 W/mK in Yuan et al., 2020), increasing the k_s is less effective for reducing T_{max}, This is because that the substrate thermal resistance (R_s) significantly reduces to a very low value, when k_s increases to ~400 W/mK, then becomes negligible after k_s continuously increases. The model results also convey that rather than using ultra-high thermal conductivity diamond (k > 1200 W/mK), other common high thermal conductivity materials, such as AlN, SiC, or copper (300–550 W/mK), may be sufficient for bottom side cooling.

Fig. 10B illustrates the topside cooling scheme using the interconnection structure that the source, drain, and gate contact pads are connected onto a high thermal conductivity substrate via microbump joints, with a polymer-based underfill material for encapsulation. In such scheme, solder bump thermal conductivity (k_b), underfill thermal conductivity (k_f), and substrate thermal conductivity (k_s) affect the heat dissipation, thus were studied (Chatterjee et al., 2019; Yuan et al., 2020). T_{max} is significantly affected by the k_b or k_f. For example, T_{max} shows a ~20% reduction for k_b increasing from 50 (common solder materials) to 200 W/mK (sintered silver), and ~40% reduction for k_f increasing from 0.2 to 10 W/mK (Yuan et al., 2020). On the other hand, a ~70% reduction in total device resistance was reported in Chatterjee et al. (2019) when apply a 14 W/mK underfill. The great impact of k_b and k_f is because that solder bump and underfill are close to the heat generation region, playing the important role for heat dissipation. In terms of substrate effect, after k_s exceeds 400 W/mK, increasing the k_s is less effective for reducing T_{max}, which is also seen in the bottom side cooling scheme. A double side cooling scheme utilizes both upper and lower thermal paths to extract the heat. Thus, the thermal dissipation ability is better than bottom side cooling and top side cooling alone. The effect of each variable on channel temperature follows the similar trends observed in the bottom side and top side cooling schemes.

Chatterjee et al. (2019) also evaluated some active cooling approaches, such as Si-based microchannel and air-jet impingent. Specific geometries of the cooling modules were considered in the models. For all studied cases, the device peak temperature exceeds the stipulated limit ($T_{max} < 200$ °C). However, once the cooling modules (configuration, coolant, flow rate, etc.) are optimized, it is possible to achieve higher heat transfer coefficients (HTCs) and lower peak temperatures. Instead of depicting the complex geometry of the convective cooling modules, Yuan et al. (2020) studied the convective cooling effects by assuming a uniform HTC covering the whole surface of the substrate. T_{max} as a function of a broad range of HTC (from 10 to 10^5 W/m^2K covering the typical HTC values of air, active

single-phase and two-phase cooling methods) were modeled, providing a guideline for the selection of heatsink methods.

In addition to the basic cooling strategies, some advanced device-level thermal management strategies were proposed in studies (Chatterjee et al., 2019; Pomeroy et al., 2018; Yuan et al., 2020). A strategy that was also inherited from GaN HEMTs cooling was highly appreciated: applying the high thermal conductive dielectric material near the active region to enhance the surface lateral heat spreading. Study in Pomeroy et al. (2018) shows that replacing the gate dielectric SiO_2 layer in the MOSFET with a 400 nm-thick diamond film, using a thermal conductivity value of 200 W/mK, is predicted to result in an 18% resistance reduction, while bringing the gold metal pads in contact with a 1000 W/mK diamond heat spreader would result in a 44% reduction in device resistance. In Chatterjee et al. (2019), an 8% T_{max} reduction was achieved for the MOSFET when a nanocrystal diamond passivation with 400 W/mK was applied, whereas with the similar scheme applied for the MESFET as shown in Fig. 11A, a 15% reduction in T_{max} can be achieved using a 1 μm thick heat spreading layer with the thermal conductivity of 10 W/mK, and an even larger reduction (50%) can be achieved with a 500 W/mK heat spreader (Yuan et al., 2020). Study in Yuan et al. (2020) further shows that when the heat spreader strategy is incorporated into the double side cooling scheme (shown in Fig. 11B), the benefit is further enhanced (~35% and ~63% reductions in T_{max} for k = 10 W/mK and k = 500 W/mK heat spreader). In those studies (Chatterjee et al., 2019; Pomeroy et al., 2018; Yuan et al., 2020), the

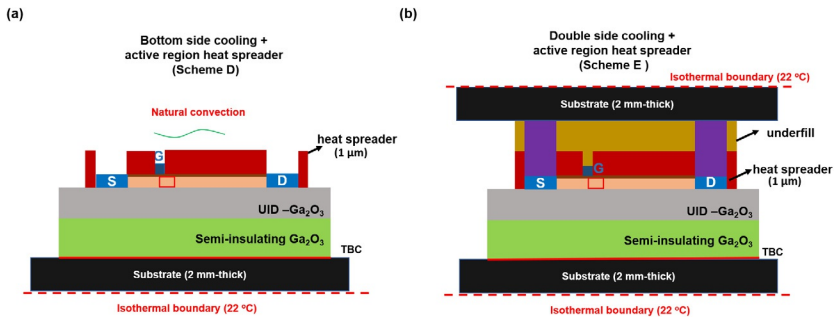


Fig. 11 The schematic of (A) bottom side cooling with the active region heat spreader scheme and (B) double side cooling with the active region heat spreader scheme. *Reproduction permissions from Yuan, C. et al., 2020. Modeling and analysis for thermal management in gallium oxide field-effect transistors. J. Appl. Phys. 127, 154502, https://doi.org/10.1063/1.5141332.*

different quantity of benefit achieved from the similar heat spreader scheme is due to not only the differences in the investigated devices structures, but also the differences in other factors such as the incorporated basic cooling schemes (i.e., bottom side cooling vs double side cooling) and the assumed thermal conductivity of the heat spreaders.

Most of the thermal management studies have focused on using the electrical-thermal simulations to evaluate various device-level cooling methods and provide a guidance for the cooling optimization. Seldom of those methods have been achieved in practice, mainly due to the technique challenges. One recent experimental work (Noh et al., 2019) has attempted the heterogeneous integrating cooling method, i.e., bottom side cooling, through transfer technique. This approach takes advantage of the transferrable property of β-Ga_2O_3 that it could be cleaved along the [100] direction. A thinned (100) β-Ga_2O_3 membrane (50–150 nm) device transferred on high thermal conductivity diamond carrier exhibit a 60% reduction in channel temperature, compared to the device on a low thermally conductivity sapphire substrate. Should not that this method is only limited to (100) β-Ga_2O_3 membrane devices, and not able to be scaled up for wafer-scale fabrications. In addition, as discussed in Section 1, the TBC is typically small (on the order of 10–20 MW/m^2K) for this van der Waals bonded β-Ga_2O_3/diamond interface. Thus, an improved integrating scheme may contribute a further reduction in device thermal resistance, resulting in an even lower device temperature. Recently, the heterogeneously integrated Ga_2O_3-on-diamond material processed by atomic layer deposition was reported with a much improved TBC (179 MW/m^2K) and an extraordinary substrate thermal conductivity (2000 W/mK) (Cheng et al., 2020b). This material can be used as a transition layer for further growth of high quality β-Ga_2O_3 buffer and devices. Thus, this heterogeneous integrating process demonstrates a promising way to delivery low thermal resistance in β-Ga_2O_3 lateral devices, ensuring the device works with great capability and reliability.

Much attention has been paid to the thermal management of β-Ga_2O_3 lateral devices, however, few studies have focused on vertical devices. For applications demanding high voltage and high power levels, vertical enhancement-mode transistors are required since they allow for superior field termination and current drive at the device level while enabling fail-safe operation and simplified designs at the system level (Cabaleiro et al., 2015; Chabak et al., 2016; Hu et al., 2019; Park and Hong, 2019; Wong et al., 2017a; Yang et al., 2018; Zhou et al., 2017b).

Average breakdown voltage of lateral devices is ~600 V, while highest reported value is 2.32 kV (Mun et al., 2019), whereas the vertical devices easily exceeds 1 kV (Hu et al., 2018a,b; Wong et al., 2017b, 2018). Unlike lateral devices, vertical devices have limitation on thermal resistances that thickness of drift layer of Ga_2O_3 cannot be decreased. However, thermal management technique of lateral devices could be still applied to vertical devices and have similar affects. Bottom side cooling with heat spreader with thermal conductivity higher than 50 W/m-K, the channel temperature will drop 30–40% compared to using 10–20 W/m-K die attach material. Also, if we could add top side cooling with underfill material with another die attach material, channel temperature will drop 50–65% compared to bottom side cooling.

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